MIDAZOLAM

High-alert drug

Check strength — 1, 2, 5 mg/mL

CHEMICAL NAME 8-chloro-6-(2-fluorophenyl)-1-methyl-4H-imidazo[1,5-a][1,4]benzodiazepine (hydrochloride or maleate)

CLASS Short-acting water-soluble benzodiazepine

TRADE NAMES Hypnovel · Dormicum · Buccolam (buccal) · Versed

THERAPEUTIC USES / INDICATIONS

- Procedural (conscious) sedation (endoscopy, minor procedures).
- Prolonged seizures / status epilepticus (buccal, intranasal, IM, IV).
- ICU sedation (infusion); premedication; induction of anaesthesia.
- Palliative care: terminal agitation, seizures (SC syringe driver).

ADMINISTRATION

- **IV** titrated slowly (e.g. 1–2.5 mg over 2 min, wait 2 min); use lower-strength (1 mg/mL) for sedation.
- **Buccal:** between gum and cheek (Buccolam prefilled syringes by age).
- **IM**, **intranasal**, **SC** infusion.
- Resuscitation equipment, oxygen, monitoring and flumazenil available.

DOSAGES

- **Conscious sedation (adult):** 2–2.5 mg IV, then 1 mg increments; usual total 3.5–7.5 mg (elderly 0.5–1 mg, max 3.5 mg).
- **Seizures (buccal):** adult/≥10 yr 10 mg; 5–10 yr 7.5 mg; 1–5 yr 5 mg; 3 months–1 yr 2.5 mg.
- **Status (IM):** 10 mg (>40 kg).
- **ICU infusion:** 0.03–0.2 mg/kg/h.
- **Palliative:** 10–60 mg/24 h SC.

PHARMACOKINETICS

- **Onset:** IV 1–3 min; buccal/IM 5–10 min. **Duration:** 1–2 h (single dose).
- Metabolised by CYP3A4 to active α-hydroxymidazolam (accumulates in renal failure).
- **t½:** 1.5–3 h (prolonged in elderly, obese, liver disease, ICU).

PHARMACODYNAMICS

- Potent **GABA-A** modulator → sedation, anxiolysis, anterograde amnesia, anticonvulsant.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Drowsiness, amnesia.
- Hiccups, nausea.
- Hypotension.

ADVERSE EFFECTS (SERIOUS)

- Respiratory depression and apnoea (especially with opioids, elderly, rapid injection); deaths from high-strength errors.
- Paradoxical agitation.
- Accumulation and delayed waking after infusions; withdrawal after prolonged infusion.

CONTRAINDICATIONS & CAUTIONS

- Severe respiratory failure; acute narrow-angle glaucoma.
- Myasthenia gravis (relative); shock.
- Use with strong CYP3A4 inhibitors (oral midazolam).

KEY INTERACTIONS

- **Opioids:** synergistic respiratory depression — reduce doses.
- **CYP3A4 inhibitors (clarithromycin, erythromycin, azoles, ritonavir, diltiazem):** ↑↑ levels.
- **Rifampicin, carbamazepine:** ↓ effect.

NURSING MANAGEMENT

- Confirm ampoule strength (high-strength 5 mg/mL restricted to anaesthesia/ICU/palliative).
- Continuous SpO_2, RR, BP, sedation score; capnography for deep sedation.
- Monitor until fully recovered; escort home; no driving 24 h.
- Teach families buccal administration for prolonged seizures (>5 min).

PATIENT EDUCATION

- You may not remember the procedure.
- Do not drive, sign documents or drink alcohol for 24 h.
- Buccal seizures rescue: give after 5 min of seizure; call ambulance if seizure continues 10 min after.

KEY POINTS

- Short-acting benzo — sedation and seizure rescue.
- Titrate slowly; check strength (1 mg/mL for sedation).
- Buccal 10 mg adult for seizures.
- CYP3A4 interactions; flumazenil reverses.

SOURCES CHECKED **[753]** Midazolam 1 mg/ml solution for injection/infusion (UK SmPC) **[754]** Midazolam, IM, IV (US label) **[755]** Midazolam, IM, IV (US label, Hikma Pharmaceuticals USA Inc)

Full references p. 432–457